

Polarization in SEDust: Implementation Notes

Abstract

These are maintenance notes for the polarization capability added to SEDust: where the polarized optics come from, how they are assembled, which implementation choices were made and why, and how far the numbers have been checked. The reader is assumed to be someone extending or repairing this code, not someone calling it. The calling interface is documented in `SEDust_user_manual.pdf`; the underlying physics is documented in `aligned_grain_polarization.pdf`. Neither is repeated here.

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1. Scope, and what lives elsewhere

Three documents cover this work and they do not overlap (Table 1). This one answers only the question *how was this built and why that way*. Where physics is needed to justify a coding choice, the relevant section of the theory report is cited rather than restated.

Question	Document
Why polarization at all; alignment theory; the polarized transfer equation; sign and index conventions; validation targets	aligned_grain_polarization.pdf (§1–§5, §9–§11)
How an RT code calls the library: signatures, units, what is already integrated	SEDust_user_manual.pdf, §5
How the code is built, why each choice was made, what was measured	this note

Table 1: Division of the polarization documentation.

2. The data source

Everything polarized derives from one file, the orientation-resolved Draine & Hensley (2021) astrodrust spheroid table:

```
data/dielectric/q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz
```

(porosity $P = 0.20$, iron fraction $f_{\text{Fe}} = 0$, axial ratio $b/a = 1.400$). Its layout is not self-describing, so it is recorded here (Table 2).

Property	Value
header	12 lines, skipped
payload order	((Q(jw,jr,jori), jw=0,1128), jr=0,168), jori=1,3
repeated	three times: Q_{ext} , then Q_{abs} , then Q_{sca}
one record on disk	the 169 sizes of one ($jw, jori$) pair
records read	$3 \text{ quantities} \times 3 \text{ orientations} \times 1129$
total values	$3 \times 3 \times 1129 \times 169 = 1,717,209$
wavelength axis	data/dielectric/DH21_wave, 1129 nodes
size axis	data/dielectric/DH21_aeff, 169 nodes
axis file format	two title lines, then the values free-format

Table 2: Layout of the orientation-resolved Q table. The wavelength and size axes are *not* parsed out of the header; they come from the companion grid files, which list the nodes the table was computed on.

Orientation index. The convention is the trap most likely to bite a future reader, and it is worth stating in full (Table 3); see also §3.6 of the theory report.

jori	geometry	meaning
1	$\mathbf{k} \parallel \hat{a}$	propagation along the symmetry axis
2	$\mathbf{k} \perp \hat{a}, \mathbf{E} \parallel \hat{a}$	field along the symmetry axis
3	$\mathbf{k} \perp \hat{a}, \mathbf{E} \perp \hat{a}$	field across the symmetry axis

Table 3: Orientation index of the DH21 table. $\hat{a}$ is the spheroid symmetry axis.

3. Derived quantities

For a grain whose symmetry axis lies in the plane of the sky and which is perfectly aligned, the polarization cross section is the difference the grain presents to the two linear polarizations, halved:

$$Q_{\text{pol}} = \frac{1}{2} [Q(\text{jori}=3) - Q(\text{jori}=2)], \quad (1)$$

formed separately from Q_{ext} and from Q_{abs} , giving $Q_{\text{pol,ext}}$ and $Q_{\text{pol,abs}}$. The random-orientation average of the same table is the trace,

$$Q_{\text{ran}} = \frac{1}{3} [Q_1 + Q_2 + Q_3], \quad (2)$$

which is the modified picket fence approximation (MPFA; §3.2 of the theory report). Cross sections follow from

$$C = Q \pi a_{\text{eff}}^2, \quad (3)$$

with a_{eff} converted to cm. The alignment efficiency is the Hensley & Draine (2023) Table 1 fit,

$$f_{\text{align}}(a) = \frac{f_{\text{max}}}{1 + (a_{\text{align}}/a)^\alpha}, \quad a_{\text{align}} = 0.0749 \mu\text{m}, \quad \alpha = 1.80, \quad f_{\text{max}} = 1.00. \quad (4)$$

4. Module map

Table 4 lists what each file is responsible for. Nothing else in the tree touches polarization.

5. Implementation decisions

This is the part worth reading again in six months.

5.1 The accumulation sites, and the two that hide

`sed_grain_loop` branches on `stoch_method` into four cases, and the polarized channel has to be accumulated wherever the total is. There are eight such statements (Table 5). Missing any one of them produces a polarized spectrum that is silently too low rather than an error.

The lesson. The 'draine' and 'qm' branches are OpenMP-parallel and accumulate into a thread-local `Jout_local` that is reduced at the end. The 'heuristic' branch is serial and accumulates into `Jout` directly. A survey of the accumulation sites conducted by searching for `Jout_local` therefore returns only the parallel branches, and an initial pass over this code identified four sites on exactly that basis. Since 'heuristic' is the *default* solver, that omission would have made the polarized emission come back identically zero in the default configuration while every other solver looked correct. Any future change that adds a term to the emission sum must be checked against Table 5 branch by branch, not by pattern search.

5.2 Why the 'qm' rescaling is exact

`qm_solve_grain` does not return a temperature distribution; it returns the emitted spectrum with C_{abs} already folded in. The polarized counterpart is therefore obtained by rescaling:

```
if (Cabs_pop(ii, ir) > 0.0_wp) then
  Jpol_local(ii) = Jpol_local(ii) + &
    wpol * emission_qm(ii) * (Cpol_pop(ii, ir) / Cabs_pop(ii, ir)) / &
    (4.0_wp * PI * lam(ii) * 1.0e-3_wp)
end if
```

This is not an approximation. The temperature distribution $P(T)$ is set by the *heating*, which is governed by C_{abs} and is unchanged by the question of which polarization is being read out. Only the emission readout changes, so

$$C_{\text{pol}} \int P(T) B_{\lambda}(T) dT = \frac{C_{\text{pol}}}{C_{\text{abs}}} C_{\text{abs}} \int P(T) B_{\lambda}(T) dT = \frac{C_{\text{pol}}}{C_{\text{abs}}} \text{emission_qm} \quad (5)$$

holds identically, wavelength by wavelength. Wavelengths with zero C_{abs} emit nothing and are skipped rather than divided by zero.

5.3 The $P(T)$ solver was not touched

No change was made to `calc_P`, `narrow_iterative`, or `qm_solve_grain`. This is deliberate and it is what makes stochastic heating automatically correct for polarization: at every site the same temperature weight multiplies both accumulators, one with $dn C_{\text{abs}}$ and the other with $dn f_{\text{align}} C_{\text{pol}}$. The polarized spectrum is thus the same average over the same $P(T)$, and no separate stochastic treatment of the polarized channel is needed (see §5.7 of the theory report for why this is the physically right structure).

5.4 Mixed orientation-average conventions

This is the one open physical wart, and it is documented rather than hidden.

C_{abs} in SEDust comes from our own T-matrix run, whose random-orientation average is computed exactly. C_{pol} comes from the release table, whose available average is the $1/3$ trace, Eq. (2), i.e. the MPFA. The two are different approximations to the same quantity, and the code mixes them: `build_Cpol` adds the polarized channel and deliberately does *not* recompute C_{abs} .

Table 6 gives the measured disagreement, $|Q_{\text{abs}}^{\text{trace}}/Q_{\text{abs}}^{\text{exact}} - 1|$, on the shared $(\lambda, a_{\text{eff}})$ grid.

Reading the far-infrared maximum. The 3.46% worst case occurs at $a = 4.2\,\mu\text{m}$, a size carrying $dn/n_H \sim 2.8 \times 10^{-36}$, so it is numerically irrelevant. Restricting to $a \leq 0.5\,\mu\text{m}$, which covers 99.999% of the geometric cross section, the worst case drops to 1.98%. At $a \ll \lambda$ both averages approach the same Rayleigh limit, which is why the far infrared is well behaved.

Verdict. The mixture is harmless for polarized *emission*, which is a far-infrared phenomenon. It is **not** defensible for polarized *extinction* in the ultraviolet or the optical, and column 8 spans the whole wavelength grid. The `Cpol_ext` output of `dust_extinction` is the same quantity by the same route and carries the same caveat unchanged. Anyone using either shortward of the near infrared must resolve this first. The clean resolution is to recompute C_{abs} on the MPFA convention so that both channels share one average. That changes the existing SEDs, so it breaks their byte-identity with the current reference outputs and has to be done as a separate, deliberately validated step rather than folded into this one. The size of the change is not an argument for leaving it: the correctness argument is that two channels of one calculation should not use two different orientation averages.

5.5 f_{align} is the fit, not the release column

`data/release/size_distribution.dat` also ships an alignment column. It agrees with Eq. (4) to a median ratio of 1.0000 but differs by up to 0.32% at the small end. The analytic form is the canonical one in SEDust because it is the published fit, because it is defined on any radius grid rather than only the tabulated one, and because it keeps the alignment model in a single place. Do not mix the two sources: a quantity computed half from one and half from the other is neither.

5.6 Names and surface area

Cpol was not renamed to Cpol_abs. The name appears at roughly 40 sites and is part of the public surface (`grain_pop_t%Cpol`, the optional `set_pop(..., Cpol_in=...)`). A rename would touch every one of them for no functional gain. The declaration carries an explicit note instead: `Cpol` is the absorption one, `Cpol_ext` its extinction twin.

The polarized extinction was exported as a table column first. A dedicated `dust_cpol_ext` accessor was considered at the time and rejected: `dust_lib` was then an emission-facing API with no extinction surface at all, so such an accessor would have been the module's only extinction entry point, a worse inconsistency than the inconvenience it removed. The condition attached to that decision was that `Cpol_ext` should join a full extinction API if one ever appeared. It has; see §5.7.

5.7 Why the accessor exists now

The table-only export was right for a library that emitted and did not extinguish. Rebuilding a radiative transfer code on top of SEDust changed the premise: the host would have obtained its emission from a call and its opacity from a

parsed text file, and that asymmetry is the whole problem. The file is also written on whatever wavelength grid the driver used, so a host on any other grid has to interpolate optics that the model could have evaluated exactly.

dust_extinction closes it. The model object already held everything required, so the accessor adds no physics and no new data source; it performs the same size integral the driver performs, over the populations already built. Two fields were added to grain_pop_t to make that possible: Cpol_ext, previously a module array only, and gsca, the scattering asymmetry, which the emission path never needed and which sed_init now interpolates onto the size grid beside C_{abs} and C_{sca} , from the same Q table and by the same routine the driver used. Agreement with the table is at the precision the file is written with (a median relative deviation of $\sim 7 \times 10^{-9}$ in each cross section), which is the expected result for one integral evaluated twice.

The table remains, unchanged byte for byte, and remains the right route for a tool that wants the optics without linking the library.

5.8 Compression handling, and a scratch file that leaked

The table ships compressed and Fortran cannot read a deflate stream. Adding a zlib binding was rejected as too heavy for one file. The reader instead shells out once,

```
call execute_command_line('gzip -dc "'//trim(gz_file)//'" > "'// &
                           trim(out_file)//'", exitstat=estat, cmdstat=cstat)
```

writes q_jori_scratch.dat in the working directory, reads it, and deletes it. A fresh clone works with no manual setup step, and neither a 17 MB untracked sibling in data/dielectric nor a new build dependency is created. A path not ending in .gz is opened directly.

The bug this had. Shell redirection creates the target file *before* gzip runs, so when decompression failed the early return left a zero-length q_jori_scratch.dat behind in the user's working directory. The failure path now calls cleanup_scratch before bailing out, like every other error exit. The general shape of the bug, an error path that skips the cleanup the success path performs, is worth watching for in any future edit to this reader: it has eight separate early returns.

6. Verification

6.1 Polarized extinction against the release

The comparison against the release is restricted to $\lambda \geq 0.11 \mu\text{m}$ because the reference file turns negative below that. The 0.9422% maximum deviation is not a systematic offset: it sits at $\lambda = 0.112 \mu\text{m}$, right at the sign reversal of the polarized extinction, where the quantity passes through zero and a relative deviation is not a meaningful measure. Away from that point the agreement is at the level of the median.

The 7×10^{-8} figure is the precision the ASCII output format carries, so column 8 and calc_pol_ext.x agree as closely as the files allow. They are two independent routes to the same quantity: calc_pol_ext.x works straight from qpol_ext

and the release size distribution, while column 8 goes through `sed_init`, the $\log a$ interpolation onto the SED size grid, and `build_Cpol`.

6.2 Polarized emission

Intrinsic polarization fraction `lamI_pol/lamI_total` under a Mathis ISRF at $U = 1.585$:

The mid-infrared zero is correct rather than a failure: there the emission is dominated by stochastically heated PAHs, which HD23 take to be unaligned.

19.15% versus Planck's 22% is a property of the model, not a bug. Planck measures $p_{\max} = 22.0(+3.5, -1.4)\%$ at 353 GHz. The 19.15% here is the *maximum* attainable value, obtained before any geometric factor, and every real line of sight reduces it, so the model as implemented cannot reach the observed maximum. This traces to the dust model itself: the axial ratio $b/a = 1.4$ is the minimum non-sphericity DH21b found the starlight polarization efficiency to require, and it was not tuned to maximize the submillimeter polarization fraction. Do not close the gap by adjusting f_{align} or the axial ratio without recognizing that as a change to the dust model, requiring its own revalidation against the extinction and emission constraints HD23 fit.

7. Limits and extension paths

These are limits of the published optics, not of the code.

No circular polarization or birefringence. The table header offers Q_{ext} , Q_{abs} and Q_{sca} only, none of which is a phase retardation, so C_{circ} cannot be formed and must be set to zero. Obtaining it means computing the fixed-orientation amplitude matrix ourselves. `tmatrix/src/ampld.lp.f` exists in the tree for exactly this and is currently excluded from the build.

No scattering matrix for aligned grains. The release gives the integrated Q_{sca} with no angular information, so scattering by an aligned spheroid cannot be modeled from it. Reaching optical polarized transfer would require wavelength-resolved Mueller matrices, from `ampld.lp.f`, DDSCAT, ADDA or CosTuuM, plus a matrix-based scattering treatment on the consumer side. The data volume is $\lambda \times \theta \times \text{orientation} \times 16$ components, which is the reason this has not been attempted.

Far infrared needs neither. In that regime the albedo is essentially zero, so the transfer requires only the extinction matrix and the emission vector, both fully determined by what is exported. The HAWC+ $154\mu\text{m}$ comparison lies inside this regime, so the far-infrared milestone is reachable with the present data and should be attempted before any scattering work.

Astro dust only. DL07 and Zubko have no polarized optics. `build_dl07` releases any polarized arrays left by a previous astro dust initialization, so `lamI_pol` returns zeros rather than stale values.

7.1 Cost of build_Cpol

Loading the orientation-resolved table is the most expensive part of model construction (Table 9).

The cheapest improvement, if this ever matters, is to release `qext_j`, `qabs_j` and `qsca_j` once the combinations of Eqs. (1)–(2) have been formed. Nothing downstream of `build_Cpol` reads them, and dropping them removes 13.7 MB, about two-thirds of the resident total. They are currently kept public because `calc_polext.f90` and ad-hoc checking programs read the orientation blocks directly; a release step would have to be opt-in or come after those consumers are served.

8. Reproducing the numbers

From SEDust/sed/:

```
make calc_polext.x # standalone polarized-extinction check
./calc_polext.x # -> output/polarized_extinction_ours.dat
                  # lambda, ours, reference, ratio

make calc_kext_astrodust.x # full-pipeline extinction table
./calc_kext_astrodust.x # -> ../data/kext_astrodust_MW.dat, 8 columns

make libsedust.a # library, for the emission path
```

`calc_polext.x` prints the median and maximum deviation against the release directly, restricted to $\lambda \geq 0.11 \mu\text{m}$. Column 8 of the extinction table should reproduce it to the ASCII precision quoted in Table 7; if the two disagree by more than that, the $\log a$ interpolation in `build_Cpol` or the size grid it interpolates onto is the place to look, since that is the only step separating the two routes.

For the emission fractions of Table 8, build a model with `build_astrodust`, evaluate `J_Mathis` at $U = 1.585$, and call `dust_emission` with `lamI_pol` requested; the example in §5.4 of the user manual is the complete recipe.

9. References

- Draine, B. T., & Hensley, B. S. 2021b, ApJ, 919, 65 (DH21b). 2021ApJ...919...65D
- Hensley, B. S., & Draine, B. T. 2023, ApJ, 948, 55 (HD23). 2023ApJ...948...55H
- Peest, C., et al. 2023, A&A, 673, A112. 2023A&A...673A.112P
- Planck Collaboration 2020, A&A, 641, A12. 2020A&A...641A..12P
- Reissl, S., Wolf, S., & Brauer, R. 2016, A&A, 593, A87 (POLARIS I). 2016A&A...593A..87R
- Seon, K.-I. 2018, ApJ, 862, 87. 2018ApJ...862...87S

File / entity	Responsibility
sed/src/q_table_jori.f90	The reader. Decompresses, parses the three quantity blocks, validates (monotone axes, finite values, no trailing data), forms Eqs. (1)–(2), and supplies $f_{\text{align_hd23}}(a)$, Eq. (4). Publishes $q_{\text{pol_ext}}$, $q_{\text{pol_abs}}$, $q_{\text{ran_*}}$ and the raw $q_{\text{ext_j}}/q_{\text{abs_j}}/q_{\text{sca_j}}$.
sed_astrodust.f90: build_Cpol	Interpolates Q_{pol} from the 169-node table axis onto the size-distribution grid in $\log a$, exactly as C_{abs} and C_{sca} are, multiplies by πa_{eff}^2 , and fills $f_{\text{align_ad}}$. Checks that the polarized and Q wavelength grids agree node for node.
sed_astrodust.f90: Cpol, Cpol_ext, falign_ad	Module-level arrays, (NLAM, NA) and (NA). Cpol is the <i>absorption</i> one and drives polarized emission; Cpol_ext is its extinction twin and drives dichroic extinction.
dust_model_mod.f90: grain_pop_t	Carries Cpol(:, :) and falign(:). Both unallocated is how the solver recognizes an unpolarized population. Also carries the extinction-side Cpol_ext(:, :) and gsca(:, :), read only by dust_extinction.
sed_astrodust.f90: set_pop	Optional Cpol_in, falign_in. Both must be supplied together; supplying neither leaves the population unpolarized. The extinction-side Csca_in, Cpol_ext_in and gsca_in are optional and independent; a population that omits them contributes zero to those terms of the size integral.
sed_astrodust.f90: sed_grain_loop	Optional Cpol_pop, falign_pop, Jpol, again all three or none (do_pol). Accumulates the polarized channel alongside Jout in every solver branch (§5.1).
sed_astrodust.f90: dust_emission	Optional lamI_pol, last argument. Sums Jpol over the populations that carry polarized optics and applies the same unit convention as lamI_total.
sed_astrodust.f90: dust_extinction	The extinction accessor. Integrates C_{abs} , C_{sca} , the scattering-weighted $\langle \cos \theta \rangle$ and $C_{\text{pol,ext}} f_{\text{align}}$ over the size distribution of every population, and returns them per H atom on $m\lambda_{\text{m}}$. These are the same numbers column 8 of the table carries, without the file (§5.7).
sed/src/calc_polext.f90	Standalone check: computes the polarized extinction directly from $q_{\text{pol_ext}}$ and the release size distribution, and compares against the published polarized_extinction.dat. Independent of sed_init.
sed/src/calc_kext_astrodust.f90	Writes column 8, $C_{\text{pol,ext}}/H$, of data/kext_astrodust_MW.dat as $\sum_a dn_{\text{Ad}}(a) C_{\text{pol,ext}}(\lambda, a) f_{\text{align}}(a)$.

Table 4: Responsibility of each file touched by the polarization update.

branch	context	accumulates into	pattern
'draine'	equilibrium grain	Jpol_local	thread-local
'draine'	stochastic, sum over $P(T)$	Jpol_local	thread-local
'heuristic'	equilibrium grain	Jpol	direct
'heuristic'	stochastic, sum over $P(T)$	Jpol	direct
'qm'	QM solve, rescaled	Jpol_local	thread-local
'qm'	GD fallback after QM failure	Jpol_local	thread-local
'qm'	equilibrium grain	Jpol_local	thread-local
'equil'	every grain	Jpol	direct

Table 5: Where the polarized emission is accumulated. The two boldface rows are the ones a search for the Jout_local pattern does not find.

band	median	maximum
far infrared ($\lambda > 30 \mu\text{m}$)	0.022%	3.46%
restricted to $a \leq 0.5 \mu\text{m}$		1.98%
mid infrared	0.033%	
near infrared	0.038%	
optical	0.080%	
ultraviolet	0.21%	9.53%

Table 6: Disagreement between the trace average and the exact orientation average of Q_{abs} .

Check	Result
Column 8 vs. released polarized_extinction.dat	median 0.0296%, max 0.9422%, 992 points
Column 8 vs. calc_polext.x	agree to 7×10^{-8} relative
Columns 1–7 of kext_astrodust_MW.dat	byte-identical to the pre-change file
astrodust_irem_ours_S1/S2/PAH.dat	byte-identical
dl07_sed_ours_mw31_60.dat	byte-identical
Build	zero warnings

Table 7: Verification of the polarized extinction path and regression of everything else.

λ	fraction
12 μm	0.00%
100 μm	13.67%
154 μm	17.15%
350 μm	18.79%
850 μm	19.15%
median over 300–3000 μm	19.18%

Table 8: Intrinsic polarization fraction of the emitted spectrum, before any geometric factor.

Item	Size
compressed file	4.1 MB
scratch file when expanded	17.2 MB
raw arrays qext_j, qabs_j, qsca_j	13.7 MB resident
derived qpol_*, qran_*	7.6 MB resident
time	about 1 s of the 3.1 s model build

Table 9: Cost of loading the orientation-resolved table.